

|   |           |        |     |
|---|-----------|--------|-----|
| A | Apellidos | Nombre | dni |
|   |           |        |     |

Calcula los límites:

$$\text{I)} \quad \lim_{x \rightarrow 0} \frac{\cos(2x) - \cos(x)}{\sin^2(x)}$$

$$\text{III)} \quad \lim_{x \rightarrow 2} \left( \frac{1}{x-2} - \frac{1}{|x-2|} \right)$$

$$\text{II)} \quad \lim_{x \rightarrow 1} x^{\frac{1}{x-1}}$$

$$\text{IV)} \quad \lim_{x \rightarrow -\infty} x^2 e^x$$

**Soluciones:**

$$\text{I)} \quad \lim_{x \rightarrow 0} \frac{\cos(2x) - \cos(x)}{\sin^2(x)} = -\frac{3}{2}$$

$$\text{II)} \quad \lim_{x \rightarrow 1} x^{\frac{1}{x-1}} = e$$

$$\text{III)} \quad \lim_{x \rightarrow 2} \left( \frac{1}{x-2} - \frac{1}{|x-2|} \right) = \nexists$$

$$\text{IV)} \quad \lim_{x \rightarrow -\infty} x^2 e^x = 0$$

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Calcula los límites:

$$1. \lim_{x \rightarrow \infty} \frac{\ln(x^2 + 1)}{\sqrt{x}}$$

$$3. \lim_{x \rightarrow 0} \left( \frac{1}{x} - \frac{1 + |x|}{x^2} \right)$$

$$2. \lim_{x \rightarrow \infty} (3x^2 - 1)^{\frac{1}{1 + \ln x}}$$

$$4. \lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - 3x + 2}$$

**Soluciones:**

$$1. \lim_{x \rightarrow \infty} \frac{\ln(x^2 + 1)}{\sqrt{x}} = 0$$

$$2. \lim_{x \rightarrow \infty} (3x^2 - 1)^{\frac{1}{1 + \ln x}} = e^2$$

$$3. \lim_{x \rightarrow 0} \left( \frac{1}{x} - \frac{1 + |x|}{x^2} \right) = -\infty$$

$$4. \lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - 3x + 2} = -3$$

|   |           |        |     |
|---|-----------|--------|-----|
| C | Apellidos | Nombre | dni |
|   |           |        |     |

Calcula los límites:

a)  $\lim_{x \rightarrow 0} (3x^2 + 1)^{\frac{1}{1-\ln x}}$

c)  $\lim_{x \rightarrow \infty} \frac{\ln(x^2 - 1)}{\sqrt[3]{x}}$

b)  $\lim_{x \rightarrow -1} \frac{3x^2 + x - 2}{x^2 - 2x - 3}$

d)  $\lim_{x \rightarrow 0} \left( \frac{1}{|x|} - \frac{1+x}{x^2} \right)$

**Soluciones:**

a)  $\lim_{x \rightarrow 0} (3x^2 + 1)^{\frac{1}{1-\ln x}} = 1$

b)  $\lim_{x \rightarrow -1} \frac{3x^2 + x - 2}{x^2 - 2x - 3} = \frac{5}{4}$

c)  $\lim_{x \rightarrow \infty} \frac{\ln(x^2 - 1)}{\sqrt[3]{x}} = 0$

d)  $\lim_{x \rightarrow 0} \left( \frac{1}{|x|} - \frac{1+x}{x^2} \right) = -\infty$